Read the Spectrum: Eyes Open vs. Eyes Closed in Resting EEG

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1. Question and approach

The question was what changes in the EEG spectrum when the eyes close. Both ~60 s recordings (64 channels, 160 Hz) were analysed twice: through the EEGLAB menus (channel locations from the supplied SFP file, average reference, channel spectra and maps, 1-45 Hz) and with a reproducible MNE-Python notebook. In Python, channel labels were cleaned by removing trailing dots only, and electrode positions were taken from the instructor-supplied PhysioNet_EEGMMIDB_64ch.sfp file (64 channels, standard EEGLAB 10-05 coordinates, PhysioNet channel order), and an average-referenced copy was analysed from 5 to 55 s. Welch PSD used a 4 s Hann window, 50% overlap and n_fft = 1024 (1-45 Hz). Spectrograms used a 2 s window, 75% overlap and n_fft = 512. Bands: delta 1-4, theta 4-8, alpha 8-13, beta 13-30 Hz. All 64 channels were successfully matched to the supplied montage; no unmatched channels were found.

2. Observations

2.1 Raw signal

In the 5-15 s window (Fig. 1), eyes-closed Oz shows a dense, regular oscillation of roughly 10 cycles per second with peaks of about ±100-200 µV, whereas eyes-open Oz looks more irregular with a lower-frequency, mixed character. The eyes-open Fz trace contains one large biphasic transient at about 9.6 s (about +250 µV then -200 µV), which is not seen at Cz, Pz or Oz at the same size and fits a blink or eye movement. In the eyes-closed traces all four channels share a negative stretch of about -100 to -150 µV in the first ~0.7 s of the window (around 5-5.7 s); because it is common to several channels, it looks like a slow drift or start-up transient rather than local activity.

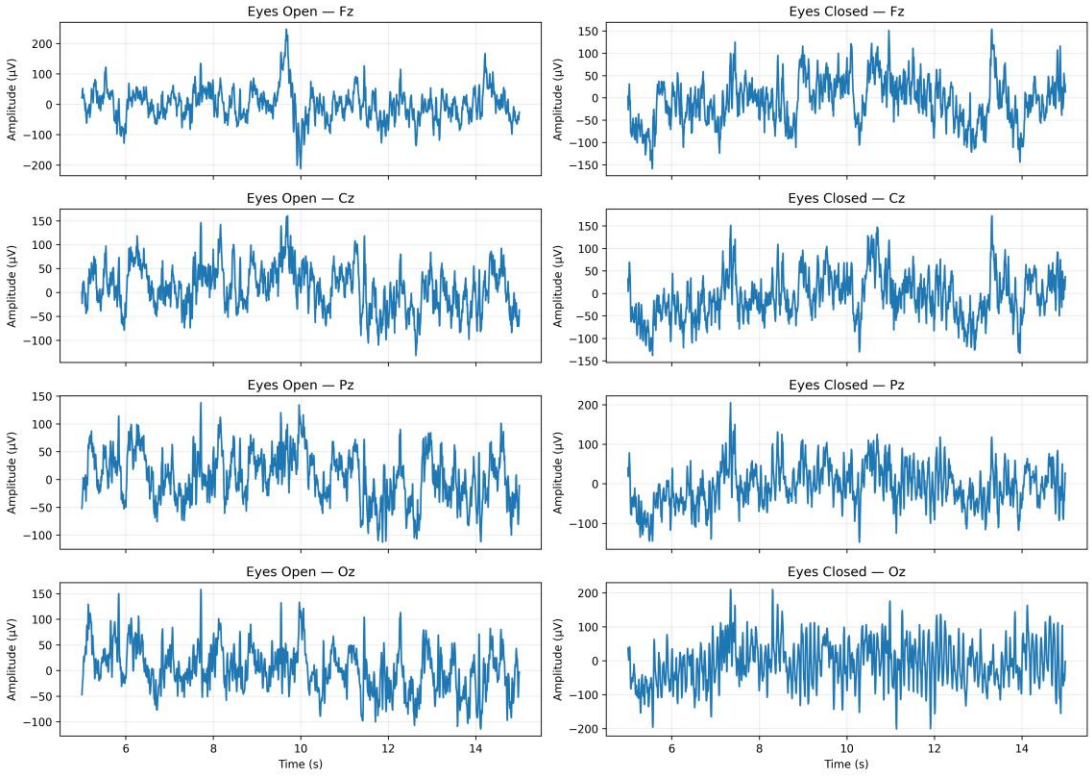


Figure 1. Raw EEG, Fz/Cz/Pz/Oz, 5-15 s, eyes open (left) and eyes closed (right). Amplitude in µV.

2.2 Power spectra

Eyes-closed Oz has a single sharp peak at 10 Hz (about 2.5e-9 V^2/Hz); the eyes-open Oz spectrum has no comparable peak and instead falls off smoothly with frequency (Fig. 2). The posterior-channel average shows the same pattern with a lower peak (about 1.4e-9 V^2/Hz at 10 Hz), as expected from averaging channels whose alpha amplitude differs. In the eyes-closed condition there is also a weak, broad elevation at roughly 15-21 Hz (Fig. 2, and the band visible in Fig. 4). Eyes-open spectra have more power below ~3 Hz.

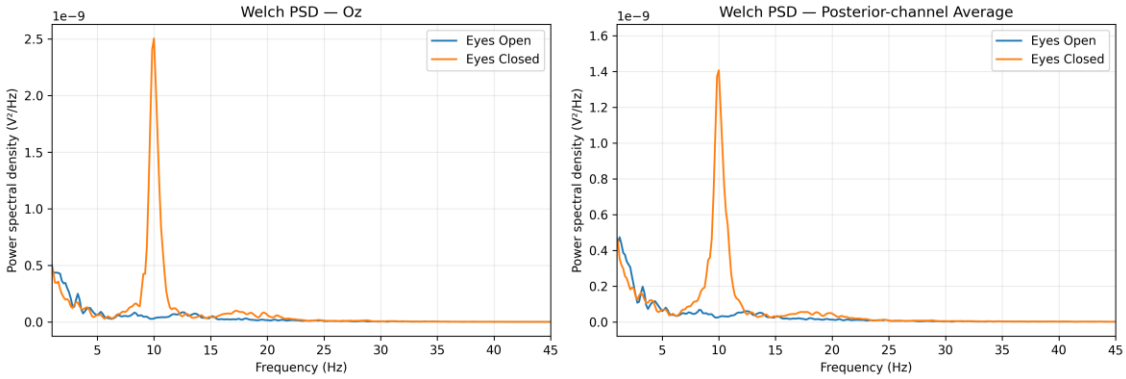


Figure 2. Welch PSD (V^2/Hz) for Oz (left) and the posterior-channel average (right), eyes open vs. eyes closed.

2.3 Quantification (results.csv)

Measure	Eyes open	Eyes closed	Closed / open
Oz alpha power (V^2)	2.63e-10	2.92e-09	11.1x
Oz relative alpha	0.166	0.664	4.0x
Posterior mean alpha power (18 channels, V^2)	2.15e-10	2.01e-09	9.4x
Posterior mean relative alpha	0.156	0.596	3.8x
Oz beta power (V^2)	3.16e-10	6.14e-10	1.9x
Oz delta power (V^2)	7.69e-10	5.95e-10	0.8x
Alpha-peak freq., posterior mean ± SD (Hz)	9.48 ± 1.87	9.92 ± 0.08	-
Alpha-peak freq., posterior range (Hz)	8.28-12.50	9.84-10.00	-

Table 1. Key features. Posterior set: P7-P8 row, Po7-Po8 row, O1, Oz, O2, Iz, Pz, Poz (18 channels).

Alpha power rose in eyes-closed in almost every channel, but by far the most at posterior sites. The largest absolute closed-minus-open differences were at Po8 (+4.04e-9 V^2), O1 (+3.35e-9), O2 (+3.23e-9), Oz (+2.65e-9) and Po4 (+2.49e-9); the largest ratios were Po8 (19.7x), O2 (13.1x), P8 (13.0x), Po4 (12.3x) and O1 (12.2x). The scalp maps (Fig. 3) show an occipito-parietal distribution with two maxima, the stronger one on the right, and almost no change frontally or at the temporal edge (T10: +2.9e-11 V^2). Eyes-open alpha power is low and spatially flat.

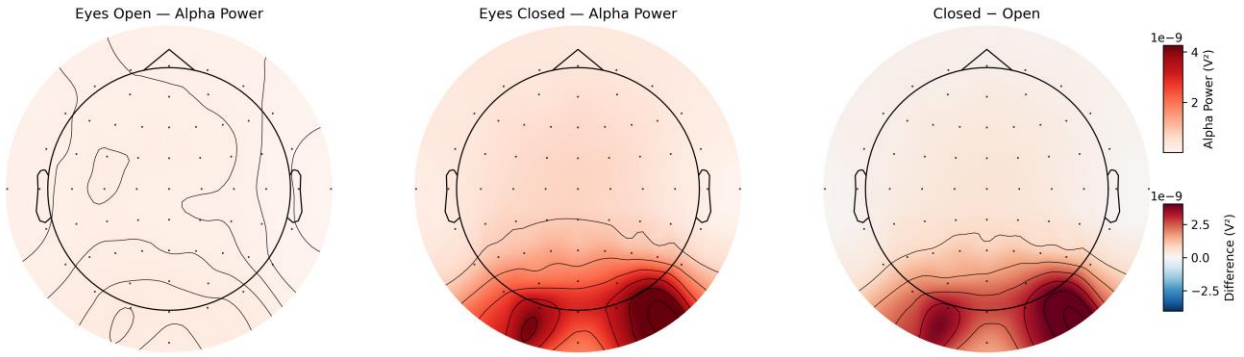


Figure 3. Alpha-band power (V²) scalp maps: eyes open, eyes closed, and closed minus open (shared colour scale for the two condition maps; the difference map has its own diverging scale).

2.4 Time-frequency structure

In the Oz spectrograms (Fig. 4), eyes-closed shows a continuous narrow band near 10 Hz for the whole 5-55 s analysis window with only small fluctuations in strength (peak about -90 dB re 1 V²/Hz), plus a fainter band at roughly 15-21 Hz. Eyes-open has no sustained band; it shows short bursts below ~3 Hz and one brief burst at about 12-15 Hz near 25 s.

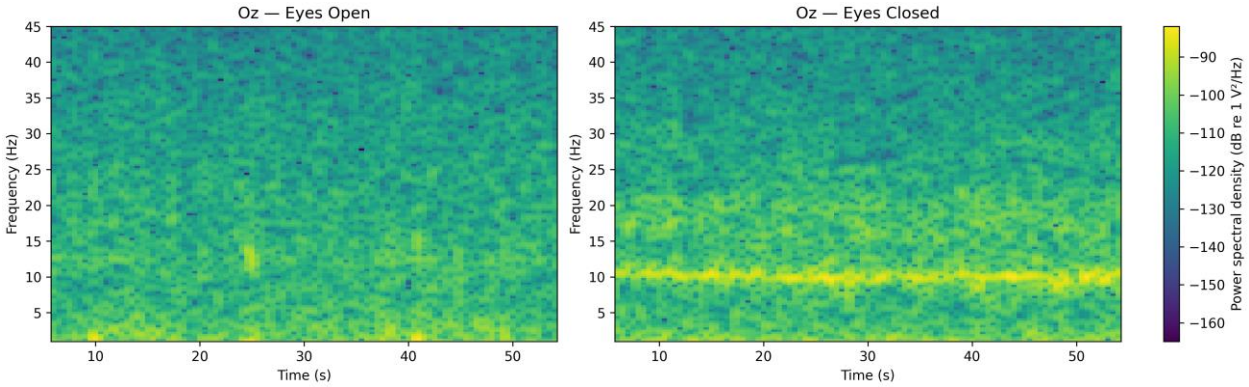


Figure 4. Oz spectrograms (2 s window, 75% overlap, n_fft = 512), 1-45 Hz, identical colour scale (dB re 1 V²/Hz).

3. Answers to the six questions

Q1. Where is the strongest difference? The strongest difference is in the alpha band, centred at 10 Hz, over occipito-parietal electrodes (Oz, O1, O2, Po4, Po8), with a right-sided maximum at Po8. At Oz alpha power is about 11 times higher with eyes closed, and relative alpha rises from 0.17 to 0.66. This is the classic posterior alpha response to eye closure (Berger effect). The right-sided asymmetry is a single-subject, single-minute observation, so I do not interpret it further.

Q2. Alpha-peak frequency and stability. Eyes closed: 10.0 Hz at Oz, and 9.92 ± 0.08 Hz across the 18 posterior channels (range 9.84-10.00 Hz, i.e. at most one or two frequency bins of 0.156 Hz), so the estimate is very stable. Eyes open: the CSV reports 12.5 Hz at Oz and 9.48 ± 1.87 Hz across posterior channels, but this is not a trustworthy alpha peak. There is no visible peak in the eyes-open PSD, the argmax simply lands on the edges of the 8-13 Hz band (12 posterior channels at 8.28-8.44 Hz, 6 at 12.34-12.50 Hz). The large spread therefore reflects the absence of a peak, not real variability of alpha frequency.

Q3. Possible non-neural peaks and shapes. No narrow line-noise peak is visible in the 1-45 Hz range in either condition. Power-line noise (50 or 60 Hz) lies outside the analysed range, so I cannot rule it out and would need to extend the PSD to the Nyquist limit to check. Eye activity: eyes-open frontal-polar delta is much higher than eyes-closed (mean over Fp1/Fp2/Fpz/Af3/Af4/Af7/Af8/Afz: 2.72e-9 vs. 7.1e-10 V²; Fp1 3.72e-9 vs. 8.8e-10), together with

the large Fz transient at ~9.6 s, consistent with blinks and eye movements. Muscle: no clear broadband rise at high frequencies is seen up to 45 Hz. The eyes-closed 15-21 Hz bump and the higher beta at occipital sites (Oz 3.2e-10 to 6.1e-10 V^2) may be a harmonic or a related rhythm of the strong 10 Hz alpha rather than a separate source; the data here cannot separate these. Recording conditions (a single 1 min session per condition, fixed run order) may also contribute to slow drifts.

Q4. Shorter Welch window. With 160 Hz sampling, a 4 s window has 640 samples and an intrinsic resolution of about 0.25 Hz; n_fft = 1024 only zero-pads this to a 0.156 Hz grid. For the 50 s analysis segment this gives 24 overlapping segments. A 2 s window would give a 0.5 Hz resolution (peaks broader and lower, nearby components harder to separate) but about 49 segments, so the PSD estimate would be smoother and less variable (less than a factor of two because overlapping segments are not independent). A shorter window therefore trades frequency resolution for lower variance.

Q5. Nyquist frequency. A signal can only be represented unambiguously up to half the sampling rate. For 160 Hz data the Nyquist frequency is 80 Hz; components above it fold back (alias) into lower frequencies if they are not removed before sampling. In practice the interpretable range is lower, and the assignment limits analysis to 45 Hz, because anti-alias filter roll-off approaches Nyquist and the highest frequencies are dominated by muscle and line noise.

Q6. EEGLAB vs. MNE. The two pipelines agree on the main findings. In both, the eyes-closed Oz spectrum has a single sharp alpha peak at about 10 Hz (EEGLAB: ≈31 dB re 1 $\mu V^2/Hz$; MNE: ≈2.5e-9 V^2/Hz), while the eyes-open spectrum has no distinct peak and declines smoothly with frequency. In both, the 10 Hz scalp map shows the eyes-closed alpha increase concentrated over occipito-parietal electrodes, with little change frontally. The visible differences are expected and do not change the conclusion. First, the units and scale differ: EEGLAB plots power in dB, $10 \cdot \log10(\mu V^2/Hz)$, whereas MNE reports linear power in V^2/Hz, so the peak looks proportionally much larger in the MNE plots and the absolute numbers cannot be compared directly. Second, the spectral estimation settings differ: MNE used the fixed Welch parameters of the assignment (4 s Hann window, 50% overlap, n_fft = 1024, 5-55 s, 1-45 Hz), while EEGLAB used its own default settings and the whole recording, which changes frequency resolution and smoothness. The EEGLAB values I read are also sampled at 1 Hz steps, so the peak frequency is less precise there. Third, the scalp maps differ in interpolation and color scaling: EEGLAB scales each map separately, whereas the MNE maps use a shared color scale. Finally, the average reference was applied in both, but the analyzed time windows were not identical. For these reasons I expect qualitative agreement, not identical numbers.

4. Limitations

This is one subject with one ~60 s recording per condition, so nothing here is a statistical test; the right-sided asymmetry and the 15-21 Hz bump are descriptive only. No artifact rejection or ICA was applied, so frontal delta and any high-frequency power include non-neural activity. Eyes-closed was always recorded second, so order and drowsiness effects cannot be separated from the condition effect. The average reference with 64 channels spreads strong occipital alpha into other channels, which can slightly alter the maps. Band power values depend on the Welch parameters fixed by the assignment.